

Assignment 1: Computational Statistics

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GitHub link: <https://github.com/Hashna86/Computational-Statistics>

Answer to the question 1

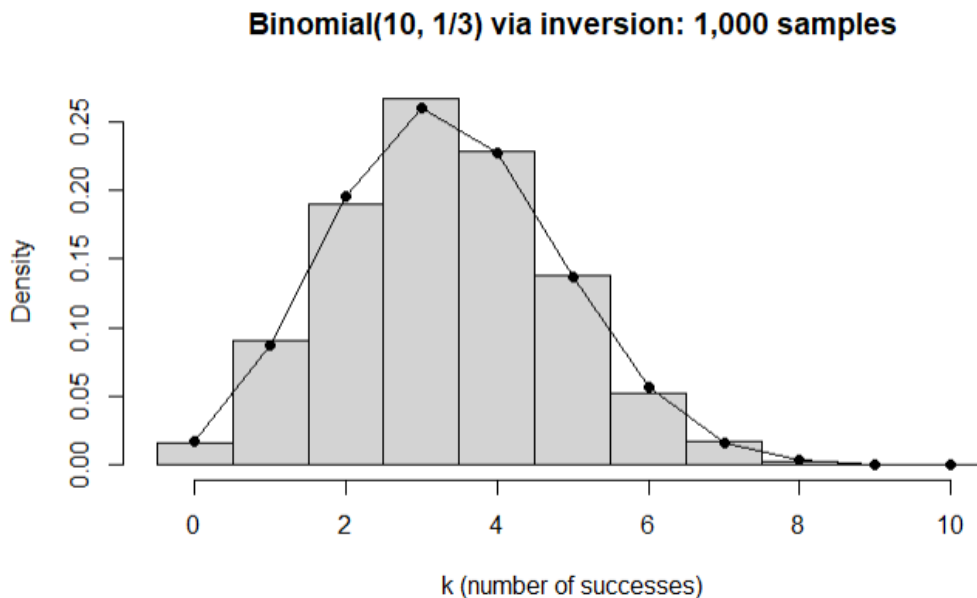
1 (a) Use the inversion method to sample from a Binomial(10, 1/3) distribution and draw histogram.

Solution:

To draw sample from a Binomial(10, 1/3) distribution by using the inversion method for a discrete distribution. Let the support be $x \in \{0, 1, \dots, 10\}$ with pmf $p(x) = \Pr(X = x)$ and CDF $F(x) = \sum_{j \leq x} p(j)$. For each draw, generate $u \sim \text{Unif}(0, 1)$ and we use pseudo-inverse of the cumulative distribution function. Then use it to draw x from our target distribution using the generalized (quantile) inverse

$$F^{-1}(u) = \inf\{x: F(x) \geq u\}.$$

Figure 1: Histogram of 1000 samples obtain from inversion transformation methods to sample from the same Binomial(10, 1/3)



1(b). Use a combination of the inversion and transformation methods to sample from the same Binomial (10, 1/3) distribution and draw histogram of 1, 000 samples.

Solution:

Here, U is the random variable(s) drawn from a Uniform (0,1) distribution that we transform into Bernoulli trials.

Let $U_{ij} \sim iidUnif(0,1)$, $p = \frac{1}{3}$, $i = 1, \dots, n$, $j = 1, \dots, 10$.

Define Bernoulli trials by inversion:

$$B_{ij} = \mathbf{1}\{U_{ij} \leq p\}.$$

Then $\Pr(B_{ij} = 1) = \Pr(U_{ij} \leq p) = p \Rightarrow B_{ij} \sim \text{Bernoulli}(p)$, independent.

Step 1 — Inversion gives Bernoulli(p)

For any i, j ,

$$P(B_{ij} = 1) = P(U_{ij} \leq p) = p, \text{ and } P(B_{ij} = 0) = 1 - p.$$

Thus $B_{ij} \sim \text{Bernoulli}(p)$. Because the U_{ij} are independent and identically distributed, the B_{ij} are also i.i.d. Bernoulli(p).

(Equivalently: the Bernoulli pseudo-inverse chooses 1 if $U \leq p$, else 0.)

Step 2 — Transformation (sum of Bernoullis is Binomial)

Let the sampled count be

$$X_i = \sum_{j=1}^{10} B_{ij}$$

For any x in $\{0, 1, \dots, 10\}$,

$$\Pr(X_i = x) = \binom{10}{x} p^x (1 - p)^{10-x},$$

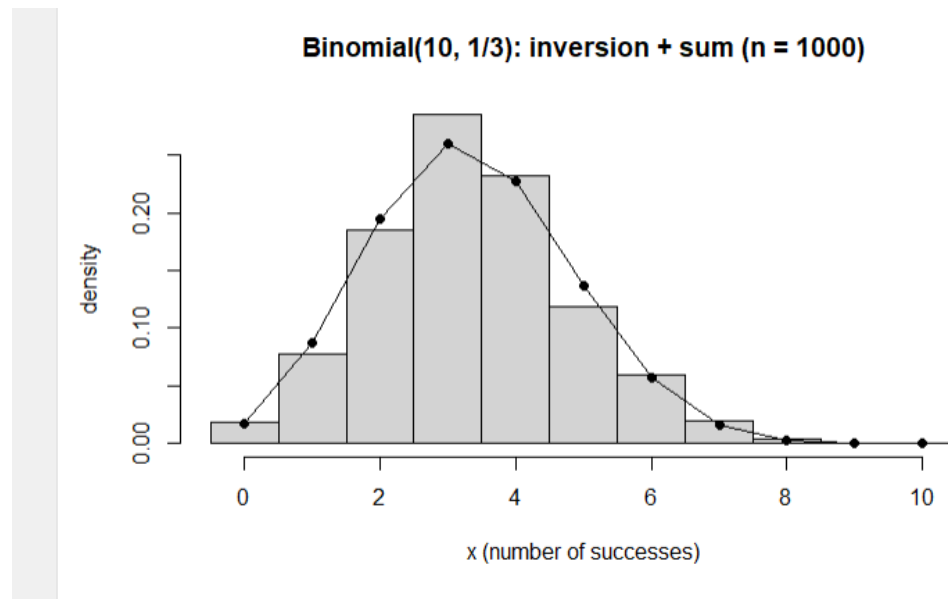
since there are $\binom{10}{x}$ ways to choose the x successes; each such outcome has probability $p^x (1 - p)^{10-x}$. Therefore $X_i \sim \text{Binomial}(10, p)$.

Conclusion

Using inversion on Uniform (0,1) to obtain i.i.d. Bernoulli(p) trials and then the transformation

$$X_i = \sum_{j=1}^{10} B_{ij} \text{ exactly } X_i \sim \text{Binomial}(10, p).$$

Figure: Histogram of 1000 samples obtain from combination of the inversion and transformation methods to sample from the same Binomial(10, 1/3)



To check correctness of our sample we compare mean and variance of sample with true value.

```
> cat(sprintf("Mean:  hat = %.3f,  true = %.3f\n", mean_hat, mean_true))
Mean:  hat = 3.353,  true = 3.333
> cat(sprintf("Var :  hat = %.3f,  true = %.3f\n", var_hat,  var_true))
Var :  hat = 2.185,  true = 2.222
\
```

Theoretically/true mean and variance of (Binomial(10, 1/3)) are

$$E[X] = np = 10 \cdot \frac{1}{3} = 3.333,$$

$$\text{Var}(X) = np(1 - p) = 10 \cdot \frac{1}{3} \cdot \frac{2}{3} = 2.222.$$

Sample mean and variance of (Binomial(10, 1/3)) for n = 1000 samples are

$$\hat{\mu} = 3.353 \text{ and } \hat{\sigma}^2 = 2.185.$$

From the above result we can say our combo samples reflects the theoretical Binomial(10, 1/3) distribution. Because here with 1000 draws we obtained $\hat{\mu} = 3.353$ and $\hat{\sigma}^2 = 2.185$ which is close to close to the theoretical mean and variance respectively 3.333 and 2.222.

1 (c) Use 100 samples generated with one of the methods above to estimate the expectation of a Binomial(10, 1/3) distribution and provide the standard error of the estimate and approximate confidence bounds as per the Central Limit Theorem.

Solution:

Now we generate 100 Binomial(10, 1/3) samples using the combo method and computes the Monte Carlo mean, its standard error, and the 95% CLT confidence interval. Here, we state those steps in short-

- 1) Draw $x_1, \dots, x_{100}$ i.i.d. from Binomial $\left(10, \frac{1}{3}\right)$.
- 2) Monte Carlo estimator of $E[X]$: $\hat{\mu} = \bar{x} = \frac{1}{100} \sum_{i=1}^{100} x_i$.
- 3) Standard error of $\hat{\mu}$ (sample sd): $se(\hat{\mu}) = \frac{s_x}{\sqrt{100}}$.
- 4) 95% CLT CI: $\bar{x} \pm z_{0.975} se(\hat{\mu})$ with $z_{0.975} \approx 1.96$.

```
> ## Print
> cat(sprintf("MC estimate E[X]: %.3f\n", mean_hat))
MC estimate E[X]: 3.260
> cat(sprintf("Standard error   : %.3f\n", se_hat))
Standard error   : 0.143
> cat(sprintf("95%% CLT CI      : [%.3f, %.3f]\n", ci_95[1], ci_95[2]))
95% CLT CI      : [2.979, 3.541]
```

From the result we found that using 100 i.i.d. samples from Binomial $\left(10, \frac{1}{3}\right)$, our Monte Carlo estimator of the mean is $\hat{\mu} = 3.260$. The estimated standard error is $\frac{s}{\sqrt{n}} = 0.143$, yielding a 95% CLT confidence interval $\hat{\mu} \pm 1.96 SE = [2.979, 3.541]$. The true value $\frac{10}{3} = 3.333$ lies inside this interval, and the deviation $|\hat{\mu} - 3.333| = 0.073$ is about **0.51 SE**, which is well within normal sampling variability for $n = 100$. As n increases, the SE (and CI width) will decrease at the usual $\frac{1}{\sqrt{n}}$ rate.

Answer to the question 2

We know that if $X \sim \text{Binomial}(n, p)$, the mean is $E[X] = n p$. As per Poisson approximation: when n is large and p is small with $\lambda = n p$ fixed then $\text{Binomial}(n, p) \approx \text{Poisson}(\lambda)$. In a Poisson process of rate 1, the count over an interval of length t is $N(t) \sim \text{Poisson}(\lambda)$ with $\lambda = t$. Hence, when relating to Binomial and Poisson process viewpoints, we match means via $n p = \lambda = t$.

Method one: Poisson process (transformation via interarrivals)

In the method unit-rate Poisson process. Interarrival times are i.i.d. $X_i \sim \text{Exp}(1)$ with CDF $F(x) = 1 - e^{-x}$ ($x \geq 0$) and inverse $F^{-1}(u) = -\log(u)$, $U \sim \text{Uniform}(0, 1)$. Define arrival times $S_n = \sum_{i=1}^n X_i$. The count up to time t is $N(t) = \max\{n : S_n \leq t\}$.

For any $n \geq 0$,

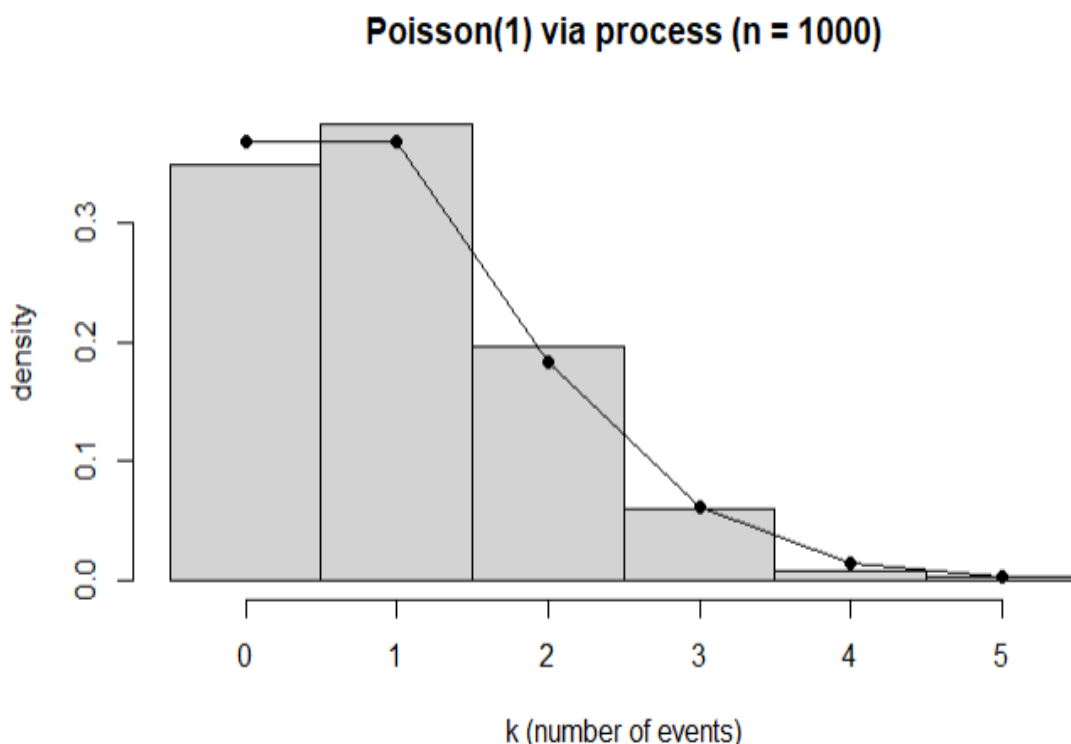
$\{N(t)=n\} \leftrightarrow \{S_n \leq t < S_{n+1}\}$. Since $S_n \sim \text{Gamma}(n, 1)$, we get the Poisson(t) pmf,

$$P(N(t)=n) = e^{-t} t^n / n!$$

Hence, $N(t) \sim \text{Poisson}(t)$ and $E[N(t)] = t$.

Then for sample $n=1000$ we draw Histogram with the Poisson (1) pmf overlaid. Also, MC mean $\hat{\mu} = \bar{x}$; SE of the mean $= \frac{s_x}{\sqrt{1000}}$; $= s_x / \sqrt{1000}$; CLT 95% CI $\bar{x} \pm \frac{1.96 s_x}{\sqrt{1000}}$. The true mean is 1 and should lie inside the CI.

Figure: Histogram of 1000 samples obtain from Poisson process (transformation via interarrivals)



Mean, SE and CI:

```
> cat("Method one (process):\n")
Method one (process):
> cat(sprintf("  mean_hat = %.3f\n", mean_hat))
  mean_hat = 1.006
> cat(sprintf("  SE          = %.3f\n", se_hat))
  SE          = 0.030
> cat(sprintf("  95%% CI    = [%.3f, %.3f]\n\n", ci_95[1], ci_95[2]))
  95% CI    = [0.947, 1.065]
```

Method two: Direct inverse CDF for the Poisson (discrete pseudo-inverse)

Use one uniform $U \sim \text{Unif}(0,1)$ and the **discrete** inverse-CDF rule:

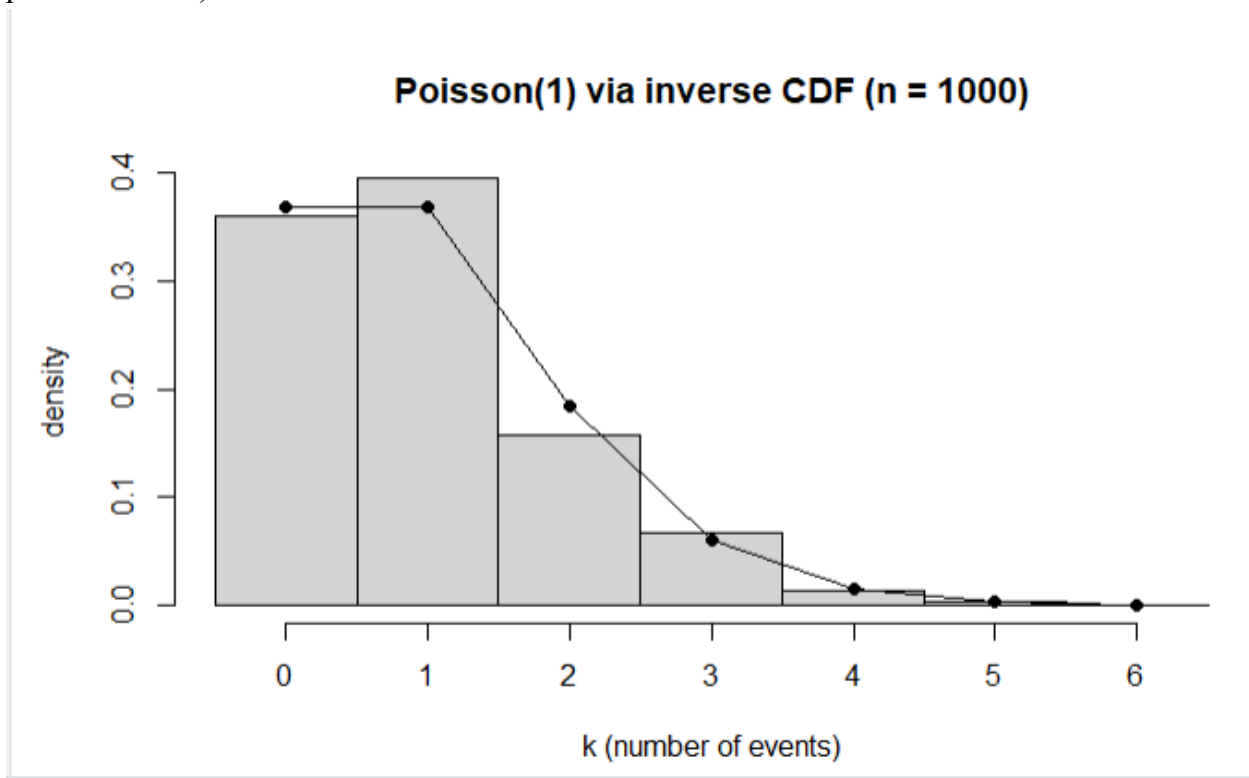
$$F^{-1}(u) = \min\{k: F(k) \geq u\}, F(k) = \sum_{j=0}^k e^{-t} \frac{t^j}{j!}.$$

Rather than recomputing factorials, update the pmf recursively:

$$p_0 = e^{-t}, p_k = p_{k-1} \cdot \frac{t}{k}; (k \geq 1).$$

This is the standard generalized inverse $F^{-1}(U)$ for a discrete law: it picks the smallest k with cumulative probability exceeding U . Because the pmf/cdf are those of $\text{Poisson}(t)$, the output distribution is exactly $\text{Poisson}(t)$. Now for sample $n=1000$ we plot the histogram with the $\text{Poisson}(1)$ pmf overlaid and also calculate MC mean, SE, and 95% CLT CI.

Figure: Histogram of 1000 samples obtain from Direct inverse CDF for the Poisson (discrete pseudo-inverse)



And calculate Mean , SE and CI:

```
> cat("Method two (inverse CDF):\n")
Method two (inverse CDF):
> cat(sprintf("  mean_hat = %.3f\n", mean_hat))
  mean_hat = 0.991
> cat(sprintf("  SE          = %.3f\n", se_hat))
  SE          = 0.032
> cat(sprintf("  95%% CI    = [%.3f, %.3f]\n", ci_95[1], ci_95[2]))
  95% CI    = [0.929, 1.053]
>
```